Substituting the column totals in the respective places in the of the estimates $\hat{a}$ and $\hat{b}$, their values can be calculated as follows:

$$\hat{b} = \frac{(9\times894.97)-(62.1\times121)}{(9\times468.97)-(62.1)^2}$$

$$= \frac{8054.73-7514}{4220.73-3856.41}$$

$$= \frac{540.73}{364.32}$$

Thus, $\hat{b} = 1.48$.

Now $\hat{a}$ can be calculated using $\hat{b}$ as

$$\hat{a} = \frac{121}{9} - \left(1.48\times\frac{62.1}{9}\right)$$

= 13.40 – 10.21

Hence, $\hat{a}$ = 3.19

Therefore, the required simple linear regression equation fitted to the given data is

$$\hat{Y} = 3.19 + 1.48x$$

It should be noted that the value of Y can be estimated using the above fitted equation for the values of x in its range *i.e.*, 3.6 to 10.7.

In the estimated simple linear regression equation of Y on X

$$\hat{Y} = \hat{a} + \hat{b}x$$

we can substitute the estimate $\hat{a} = \overline{y} - \hat{b}\overline{x}$. Then, the regression equation will become as

$$\hat{Y} = \overline{y} - \hat{b}\overline{x} + \hat{b}x$$

$$\hat{Y} - \overline{y} = \hat{b}(x - \overline{x})$$

It shows that the simple linear regression equation of Y on X has the slope $\hat{b}$ and the corresponding straight line passes through the point of averages $(\overline{x}, \overline{y})$. The above representation of straight line is popularly known in the field of Coordinate Geometry as 'Slope-Point form'. The above form can be applied in fitting the regression equation for given regression coefficient $\hat{b}$ and the averages $\overline{x}$ and $\overline{y}$.

As mentioned in Section 5.3, there may be two simple linear regression equations for each X and Y. Since the regression coefficients of these regression equations are different, it is essential to distinguish the coefficients with different symbols. The regression coefficient of the simple linear regression equation of Y on X may be denoted as b_{YX} and the regression coefficient of the simple linear regression equation of X on Y may be denoted as b_{XY}.

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